

Assignment 1

14X062 Computational Imaging

Task 1: image filtering

Based on the selected image (preferably never compressed and stored in lossless compression format) to develop the image filtering using convolutions implemented with both spatial-domain and Fourier-domain computations. Use the commands `scipy.signal.convolve2d`, `pypher.psf2otf`, `numpy.fft.fft2` and `numpy.fft.ifft2`.

- 1) Implement low-pass filtering for three Gaussian convolution kernels with the following standard deviations: 0.1, 1, and 10. Use the function `fspecial_gaussian_2d` to generate the blur kernels.
- 2) Implement high-pass filtering for three different kernels. Use the same kernel size of 101×101 pixels for each of them.

Task 2: image inverse filtering

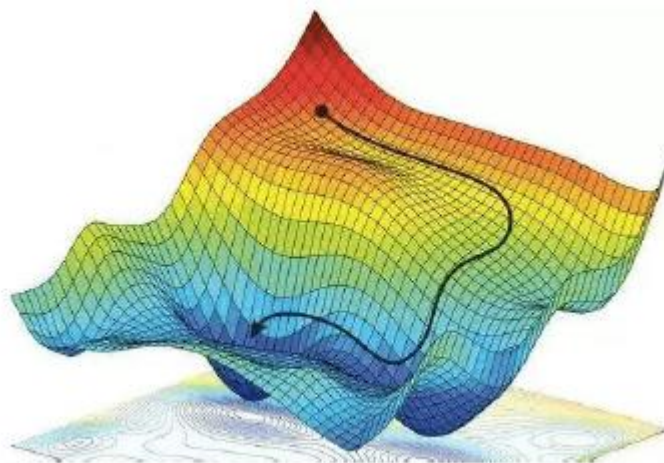
Implement inverse filtering and Wiener deconvolution. Blur the image with a Gaussian PSF with a standard deviation of 5. For the same blur kernel but add additive Gaussian noise with the four values of standard deviations (0, 0.001, 0.01, 0.1) to the blurred image before it is inversely filtered:

- 1) Inverse filtering using division in the Fourier domain.
- 2) Wiener deconvolution.

Plot the resulting images and indicate MSE/PSNR for each result.

Task 3: gradient methods

Implement gradient descent and stochastic gradient descent to solve a linear inverse problem of the form $Ax = b$.



- 1) Implement the gradient descent (GD) to find $\arg\min_x \frac{1}{2} \|Ax - b\|_2^2$. You can use the default gradient descent step size and number of iterations.
- 2) Implement the stochastic gradient descent (SGD). For each iteration of SGD, you will sample a fixed number of random rows from A (and corresponding rows from b) and then use the resulting matrix to calculate the gradient descent update. The number of rows sampled from A is the “batch size” in SGD optimization. Use the same step size and number of optimization steps as for GD. Repeat experiment with batch sizes of 10, 100, and 1000.

Plot the resulting images and indicate MSE/PSNR for each result.

To complete all results (report together with your code) into [archive](#) and submit it to Computational imaging course moodle.